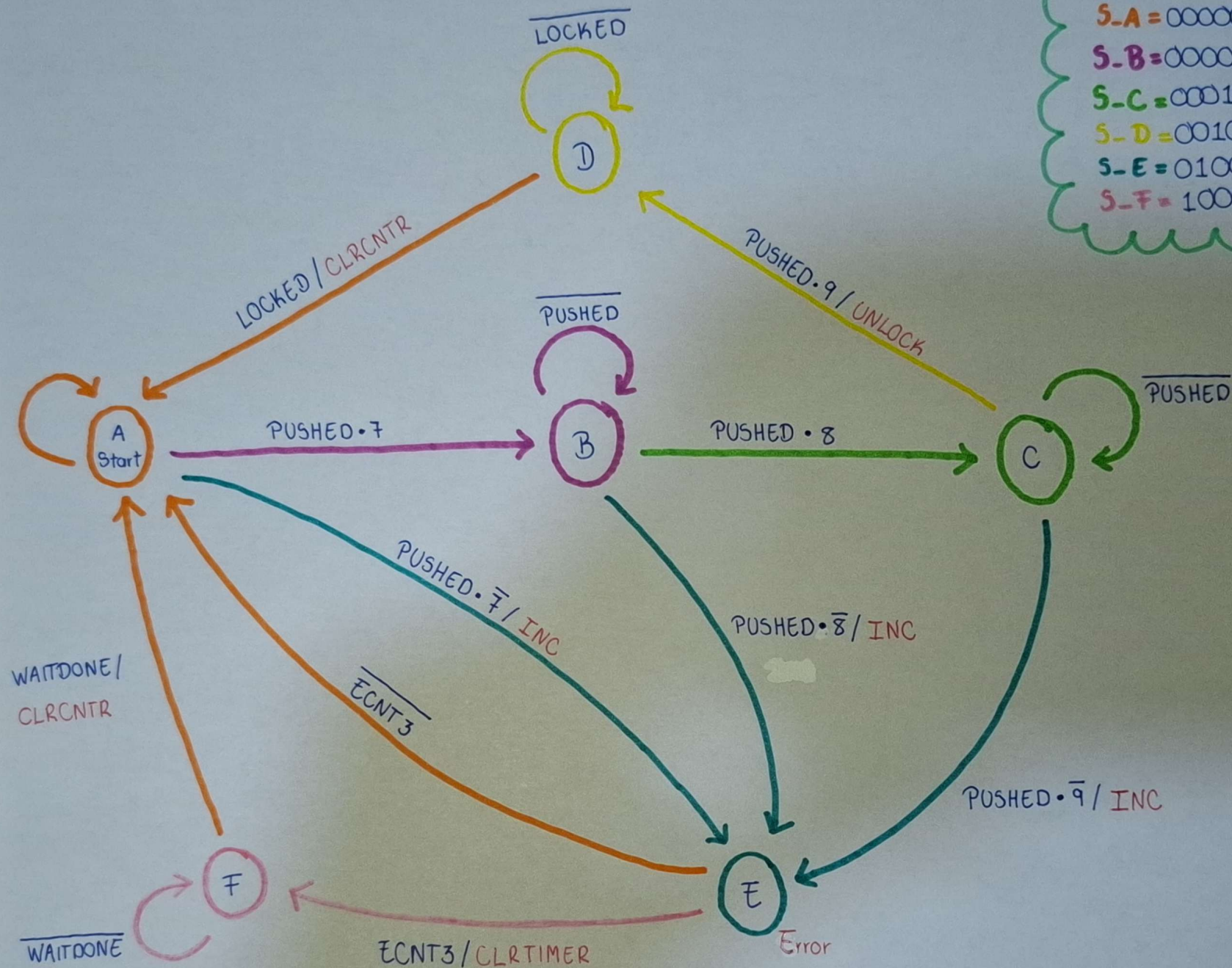


Tarea Extra - Electronic Lock

Codificación One-Hot

S-A = 000001
S-B = 000010
S-C = 000100
S-D = 001000
S-E = 010000
S-F = 100000



Lógica de siguiente estado

$$S_A = (A_start \cdot \overline{PUSHED}) + (D \cdot LOCKED) + (E \cdot \overline{ECNT3}) + (F \cdot WAITDONE)$$

$$S_B = (B \cdot \overline{PUSHED}) + (A \cdot PUSHED \cdot 7)$$

$$S_C = (C \cdot \overline{PUSHED}) + (B \cdot PUSHED \cdot 8)$$

$$S_D = (D \cdot \overline{LOCKED}) + (C \cdot PUSHED \cdot 9)$$

$$S_E = (A \cdot PUSHED \cdot \overline{7}) + (B \cdot PUSHED \cdot \overline{8}) + (C \cdot PUSHED \cdot \overline{9})$$

$$S_F = (F \cdot \overline{WAITDONE}) + (E \cdot ECNT3)$$

Lógica de salida

$$Error = E$$

$$INC = (A \cdot PUSHED \cdot \overline{7}) + (B \cdot PUSHED \cdot \overline{8}) + (C \cdot PUSHED \cdot \overline{9})$$

$$UNLOCK = C \cdot PUSHED \cdot 9$$

$$CLRTIMER = E \cdot ECNT3$$

$$CLR CNTR = (D \cdot LOCKED) + (F \cdot WAITDONE)$$